

**WAR AND PEACE STUDIES CONGRESS
(WAPSCON-2025) 21ST CENTURY ARMED CONFLICTS:
Changing Dynamics and Emerging Lessons
(23-24 October 2025-Online) (Book of Proceedings)**

Held on 23-24 October 2025, the War and Peace Studies Congress (WAPSCON-2025) brought together distinguished scholars and practitioners to analyse the evolving nature of contemporary warfare. Over the course of two days, the congress featured 45 presentations across 10 sessions, fostering a deep exploration of strategic, operational, and tactical realities in modern combat.

The discussions centered on critical themes, including intelligence and surveillance, command-and-control systems, urban warfare, and the integration of artificial intelligence and robotic systems. Beyond technological and tactical dimensions, the congress also addressed the complexities of naval and air operations, logistics, human security, and humanitarian law.

This book of proceedings compiles 23 selected articles that reflect the multidisciplinary breadth of the congress. From the geopolitical implications of emerging technologies to the intricacies of special operations and civil protection, these contributions offer rigorous academic insights into the changing dynamics of 21st-century armed conflicts. It serves as a vital resource for policymakers, military professionals, and researchers dedicated to understanding the lessons learned from the front lines of modern defence.

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**HARP
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WAR AND PEACE STUDIES CONGRESS

(E) Tuğgeneral Prof. Dr. Oktay Bingöl
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Dr. Öğr. Üyesi Selma Şekercioğlu Bozacıoğlu

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İÇİNDEKİLER

HARP SANAT YAYINEVİ YAYIN ESASLARI.....	9
PREFACE	11
CONGRESS PROGRAMME.....	12
ORGANIZATION COMMITTEE.....	21
SCIENTIFIC ADVISORY COMMITTEE.....	22
CONGRESS CHAIRPERSON OPENING REMARKS.....	27
(RTD) GENERAL ERGİN SAYGUN'S KEYNOTE SPEECH.....	31
OKTAY BİNGÖL	
PRIMARY LESSONS LEARNED FROM 21. CENTURY WARS: ACTIONS OF SELECTED COUNTRIES	37
ALİ BİLGİN VARLIK	
MILITARIZATION AND WEAPONIZATION OF SPACE: A THREAT AS- SESSMENT	55
HASAN YILMAZ	
MODERN WARFARE LESSONS FROM UKRAINE AND GAZA: DOTMLPFI IMPLICATIONS FOR FUTURE CONFLICTS.....	101
TALHA ÇAĞMAN	
THE ENDURING RELEVANCE OF CLAUSEWITZ: DEFENSE, POLITICS, AND WAR IN THE 21ST CENTURY	117
CUMHUR ERENTÜRK	
THE DIMENSIONAL CHANGES OF THE BATTLEFIELDS WITHIN THE HIS- TORICAL CONTEXT	127
AŞKIN İNCİ SÖKMEN ALACA	
HARNESSING THE POWER OF SPACE IN THE UKRAINE CONFLICT: STRA- TEGIC AND OPERATIONAL LESSONS.....	171
KOSAR KHAZAEI POUL	
THE ROLE OF AIRBORNE AND AIR ASSAULT OPERATIONS IN MODERN CONFLICTS	201

HARNESSING THE POWER OF SPACE IN THE UKRAINE CONFLICT: STRATEGIC AND OPERATIONAL LESSONS

Aşkın İnci SÖKMEN ALACA¹

Abstract

This study examines the operational and strategic roles of space capabilities in the Russia–Ukraine War (2022–present) and aims to identify key lessons from integrating space-based systems into modern military operations. The research seeks to answer how access to and integration of space assets, particularly satellite communications and commercial intelligence, surveillance, and reconnaissance (ISR), have influenced operational effectiveness, command and control continuity, and strategic resilience. Using a qualitative methodology grounded in literature review and case study analysis, the study considers the contributions of commercial space providers, including SpaceX (Starlink), Maxar Technologies, Planet Labs, and ICEYE. The findings indicate that space-based communications ensured uninterrupted command and control under contested conditions, while commercial satellite imagery enhanced situational awareness, targeting precision, and operational tempo. Furthermore, the war demonstrated that access to commercial space capabilities can serve as a force multiplier for states lacking extensive sovereign space infrastructure. At the strategic level, space capabilities contributed to national resilience, deterrence, and alliance cohesion. The study concludes that space power has become an essential component of modern warfare, fundamentally shaping operational effectiveness and strategic outcomes, and should be regarded as a critical element in future military planning and defense policy.

Key Words: Russia–Ukraine war; space operations; satellite communications; intelligence surveillance and reconnaissance (ISR); operational/strategic lessons; multi-domain operations.

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Introduction

The Ukraine–Russia War, ignited in 2022, acts as a turning point that has redefined the very fabric of modern conflict. No longer are land and air forces alone the arbiters of victory; space-based technologies have emerged as game-changers on the battlefield. Ukraine, once seen as an unlikely contender in the realm of *space power* (Hays, 2014 :57-79) in, has waged a ‘*space-enabled war*’ (Bowen, 2020; U.S. Space Force, 2020; JP 3-14, 2023) by harnessing commercial and allied assets from the US and NATO. Remarkably, Ukraine achieved this without its own satellite constellations, launch sites, or a dedicated military space force—relying instead on external command, control, and intelligence. For the first time in contemporary warfare, a nation without sovereign space might have leveraged space to challenge and diminish a stronger adversary’s capabilities. (New Geopolitics Network, 2025)

It is also significant that space assets are no longer exclusive to major powers; medium-sized states can now benefit as “user states,” defined as nations that utilize but do not own space technology. Ukraine’s advantage resulted not from the possession of space systems, but from a warfare approach that integrated “*space-derived data*”—*information from satellites and other space-based sources—into all aspects* of the conflict (Burbach, 2022). The Russia–Ukraine War has demonstrated that space-based technologies have evolved from supporting elements to critical *force multipliers* that shape the trajectory of warfare (Saric, 2022). Technologies such as commercial satellite imagery, small satellite constellations, satellite internet systems (e.g., Starlink), GNSS-based precision guidance, space-based C4ISR (Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance), and deception or countermeasure activities have been integrated into every phase of the conflict (Kostenko & Manzhula, 2025).

Ukraine has also provided real-world validation of NATO’s “*multi-domain warfare*” concept (NATO, 2020; NATO, 2022). The Alliance had developed this concept at the conceptual level for many years. Ukraine integrated its space capabilities, cyber operations, information operations, and conventional military forces into a coherent operational system. This demonstrated that cross-domain integration creates decisive advantages. By combining satellite communications (Starlink), commercial imagery intelligence (Maxar, Planet Labs), cyber defense, electronic warfare, and precision artillery, Ukraine achieved situational awareness and command agility, off-

setting Russia's numerical superiority (National Defense Lab, 2025). NATO analysts recognized that this integration exemplified multi-domain synergy, demonstrating the capacity to generate synchronized effects across all domains more rapidly than adversaries could respond. Ukraine's operational experience has informed NATO's Warfighting Capstone Concept and the Alliance's evolving multi-domain operations (MDO) doctrine. This underscores the necessity for continuous data fusion, robust public-private partnerships, and resilient space-cyber infrastructures to support deterrence and collective defense (Gilli & Gilli and Grgic, 2015:84-86)

Purpose, Scope, and Methodology of the Study

The primary purpose of this study is to analyze the role of space power at the operational and strategic levels in the context of the Russia-Ukraine war and to evaluate the lessons this war has revealed for modern military doctrine. The study seeks to answer whether space systems are merely a technical support element or a force multiplier that transforms the conduct of war. In this context, the main research question is: How has the Russia-Ukraine war revealed the role of space capabilities in operational effectiveness and strategic resilience, and how has this affected modern war doctrines? The related sub-questions are as follows: To what extent have space systems increased Ukraine's warfighting capabilities? How has the integration of commercial space actors into the war transformed the use of military force? How have space capabilities affected deterrence and the balance of power?

The study argues that space power has become a decisive force multiplier in modern warfare. This argument draws support from four sources: classical strategy theory (Clausewitz, 1976: 75-89), operational art literature (Svechin, 1992: 68-77; Isserson, 2013: 35-47), information superiority and network-centric warfare approaches (Van Creveld, 1985: 161-179; Cebrowski & Garstka, 1998: 30-33), and the views of space power theorists (Gray, 1996: 299-303; Dolman, 2002: 3-12). The example of Ukraine shows the impact of space-based communications (Starlink), commercial ISR systems (Maxar, Planet Labs, ICEYE), and navigation infrastructure on operational tempo, command, and control continuity, and targeting effectiveness. For instance, Ukrainian command and control continuity was maintained even after communications disruption, with Starlink systems enabling a reported restoration of command links within hours rather than days. Open sources suggest Ukrainian targeting cycles were reduced from an average of 48-72

hours pre-conflict to as little as 6-8 hours following the integration of commercial satellite imagery and rapid communications. Ukrainian HIMARS strikes, supported by combined satellite-GPS targeting and ISR data, are estimated in official briefings to have achieved a strike accuracy improvement of over 60 percent compared to legacy capabilities (Defense Intelligence of Ukraine, 2023). In this context, the study assesses that space systems transform not the nature of war but its character and mode of execution.

Methodologically, the study uses qualitative research design. It combines literature review, document analysis, and case study methods. Military doctrine documents and official reports serve as primary sources. Academic studies, strategic analysis reports, and dependable open-source data are secondary sources. The war in Ukraine, a unique but data-rich conflict, was analyzed using the “critical case analysis” method. This enables a comparative assessment of the operational and strategic effects of space capabilities within a theoretical framework. The study thus aims to show the role of space power in modern warfare at both conceptual and practical levels.

Ukraine’s Space Power Capability

The designation of a country as having “*space power*” means that the country not only uses space for civilian purposes but also has the capacity to access, use, and protect space, and to generate military, economic, and strategic advantages from it. Like the concepts of sea power and air power, this concept refers to the use of space for power projection. According to the US Space Force doctrine, space power is: “*A state’s capacity to operate in space, use space, and protect its national interests and project power through space systems.*” (US Space Force Doctrine, 2020: 5). According to NATO, space power is: “*The ability to support military operations, provide situational awareness, and generate operational advantages through space systems.*” (NATO, 2022: 47)

For a country to be considered a space power, it must possess five fundamental capabilities. The first and most basic criterion is the ability to access space. This includes satellite launch capability (rocket technology), independent or allied-supported launch capability, and space infrastructure. Second is satellite development capability. Communication, ISR (intelligence, surveillance, reconnaissance), navigation, and early warning satellites form the basis of military operations. The third criterion is the capacity to use space for military purposes. Military communications, targeting support, early missile warning, and operational ISR support fall within this capacity. The

fourth element is the capacity to protect space systems and ensure space security. Space power must be able to protect its own systems. Space situational awareness, anti-jamming capability, and cybersecurity fall within this protection domain. Finally, there is the ability to use space as an element of national power. Space power provides advantages in military superiority, economic power, technological superiority, and strategic deterrence. (Dolman, 2002:3-12; Bowen, 2020:115-140) According to these five criteria, countries with independent launch capabilities, military space systems, and anti-satellite systems within their own borders, such as the United States, China, and Russia, are considered full space powers; countries with regional space power, such as France, India, Japan, and Israel, which have advanced satellite systems and limited launch capabilities; and finally, developing space powers, such as Turkey, South Korea, and Brazil, which have satellite development capabilities but limited military space capabilities and no launch capabilities (Hayes, 1996).

Ukraine possessed technological expertise in space and aviation gained during the Soviet era. As a key center for rocket and space engineering, it produced the R-36 intercontinental ballistic missiles, launch vehicles, and rocket engines (Hoffmann, 2025). After independence, it was a country with a well-trained aviation and space workforce. However, it lacked its own military space program, significant satellite reconnaissance or navigation assets, launch facilities, space situational awareness (SSA), and anti-satellite (ASAT) capabilities. When assessed against the five fundamental components of space power, Ukraine has a mixed profile. Regarding the first component, access to space, Ukraine possesses significant rocket and space engineering infrastructure inherited from the Soviet era, such as the *Yuzhmash* and *Yuzhnoye* design bureaus (Pardo, 2022). However, due to post-2022 war conditions, it does not actually have an independent, sustainable launch capacity. In terms of the second component, satellite development and operation capacity, Ukraine has developed limited national ISR projects such as the Sich series of observation satellites; however, since 2022, it has largely relied on commercial Maxar, Planet Labs, ICEYE, and allied satellite systems for operational effectiveness during the war (IISS, 2025: 199). Regarding the third component, the use of space for military purposes, Ukraine has ensured command and control continuity, particularly thanks to Elon Musk's Starlink satellite communication system, and has effectively integrated space-based systems at the operational level by using satellite ISR data for preci-

sion targeting and drone operations (Watling & Reynolds, 2022 ;18–30). Regarding the fourth component, the protection of space systems and space security, Ukraine has limited independent space protection capabilities and relies heavily on allied-supported cyber and space security mechanisms. Regarding the fifth component, the use of space for national power projection, Ukraine has generated operational advantages by integrating allied and commercial space systems rather than its own national space assets, thereby creating a space-based force multiplier effect.

Within the three-tier classification of space power (full space power, regional space power, and emerging space power), Ukraine was categorized as an “emerging space power” prior to the war. Although Ukraine possessed satellite development infrastructure and space engineering capacity, it did not qualify as a full or regional space power due to its limited independent launch capability and lack of a comprehensive military space architecture. However, following the outbreak of the war in 2022, Ukraine successfully generated a de facto “*operational space power effect*” by effectively integrating allied and commercial space systems (Ogden, Knack, Lebret, Black & Mavroudis, 2024).

This demonstrates that, despite not being an independent space power in the classical sense, Ukraine developed an “integration-based space power” model capable of effectively utilizing space-based systems at the operational level. This model illustrates that, in modern warfare, space power can be generated not only through national ownership but also through allied access and commercial space integration (NATO, 2023:45-49).

When the war began, Ukraine achieved “space-enabled military effectiveness” by developing integration with commercial space companies, allied states, and the global information ecosystem to establish an asymmetric balance against Russia. Ukraine did not become a space power in the traditional sense, but it became a state that used space effectively. The three key factors underlying this transformation are (Ministry of Defense of Ukraine, 2023) as follows:

1. Communication and Imaging Capabilities Provided by Private Commercial Satellite Companies: In the field of national security, data from private commercial satellite companies can be used alongside state satellites. During times of war, military-civilian cooperation has become a strategic necessity. SpaceX, which provides satellite communication infrastructure, and

private sector companies Maxar Technologies, Planet Labs, and BlackSky, which provide satellite imaging services, have become influential enough to determine the course of the war. This shows that commercial satellite companies can be considered military strategic providers. It is known that Starlink provided command and control in Ukraine during its operational period. In providing uninterrupted information, especially photographs, it has been a strategic communication support for Ukraine against Russian disinformation and propaganda. Satellite imagery has shown Russia's buildup to the world before the war, turning global public opinion in Ukraine's favor and gaining the upper hand in the information war.

While integrating private-sector space capabilities offers significant advantages, it also creates new dependencies and operational risks. If a major commercial provider were to withdraw services mid-conflict—whether due to technical, commercial, or political reasons—the effect could be severe, potentially degrading command, control, and intelligence functions at critical moments. To mitigate such vulnerabilities, states should establish contingency plans that include developing backup communication infrastructure, diversifying suppliers, securing contractual guarantees for the continuity of service, and maintaining rapid-deployment kits for alternative networks. By foreshadowing and planning for potential provider withdrawal, military planners can strengthen operational resilience and protect against abrupt capability losses.

2. Intelligence Sharing with Allied Countries: Intelligence support provided by NATO and EU countries has created operational awareness. NATO's satellite support to Ukraine has demonstrated the importance of space data in collective defense. Without NATO satellite support, Ukraine might not have been able to withstand the war for as long as it did (NATO, 2023).

3. Flexible Command and Control Structures That Integrate Space-based Information Into Operational Decision Cycles: Satellite communications, imaging, and navigation have been integrated into areas such as rapid decision-making, target detection, unit coordination, and logistics management. Satellite internet ensured the continuity of Ukraine's command and control network even under intense electronic warfare conditions. The combination of satellite imagery, GPS, and UAVs has been found to enhance Ukraine's deep and precision-strike capability. Ukraine was able to read the battlefield with real-time satellite imagery and gained agility in troop deployment/logistics decisions.

Table 1: Space Capabilities Transforming Ukraine’s Combat Capabilities

Capability Area	Space Contribution	Impact on Warfighting Capability
Communications (C3)	Starlink satellite internet	Sustained command and coordination under attack
ISR (Intelligence, Surveillance, Reconnaissance)	Commercial and allied satellite imagery	Enhanced situational awareness and operational planning
Targeting and Fires	Satellite-guided weapons	Improved firing accuracy and operational efficiency
UAV Operations	GPS and satcom integration	Enabled real-time reconnaissance and strike capability
Information Warfare	Satellite imagery for verification	Increased global credibility and international support
Resilience	Space-based redundancy against cyber and jamming threats	Sustained operational tempo

As illustrated in the table, space-based capabilities significantly transformed Ukraine’s warfighting capacity by enhancing effectiveness across multiple operational domains. Satellite-supported communications ensured continuity of command and control under sustained attack, while commercial and allied ISR improved situational awareness and operational planning. The integration of satellite-guided weapons and GPS-enabled UAV operations increased precision, responsiveness, and strike efficiency. At the same time, the use of satellite imagery in information warfare strengthened international credibility and political support. Finally, space-enabled resilience—through redundancy against cyber and electronic warfare threats—allowed Ukraine to sustain operational tempo. Overall, the table demonstrates that space systems functioned not merely as auxiliary tools but as critical enablers, shaping the battlefield. These three factors position Ukraine as a case study in military history for effectively leveraging space data without possessing indigenous space assets. Ukraine achieved three critical advantages over Russia through space: narrowing the space power gap via commercial private-sector partnerships, accelerating the command, control, communications, and targeting cycle with space-derived data, and attaining superiority in informa-

tion warfare, consequently influencing political and strategic domains. (Pivaru, 2025) Rather than launching its own satellites, Ukraine transformed its combat capabilities by utilizing private sector and allied space systems as strategic leverage. In this context, strategic leverage refers to a state's ability to amplify its influence by utilizing resources or structures beyond its own capabilities. Support from NATO and EU countries in military, technological, political, economic, and doctrinal areas has been a key factor in sustaining Ukraine's war effort over three years.

Ukraine's use of space power reveals three fundamental truths: space is a critical determinant of modern warfare; even a state without its own satellites can gain asymmetric power in conflict; and the side that best analyzes data gains an operational advantage (Gleason, 2024). Following the analysis of Ukraine's position within the framework of space power components, it is essential to address the role of commercial space actors to understand how this capacity was produced during the war. The following sections briefly examine the Starlink system's decisive role in satellite communications and the contributions of Maxar Technologies, Planet Labs, and ICEYE to commercial satellite intelligence.

Integration of Commercial Satellite Systems into Military Operations: The Starlink Example

The Ukraine–Russia war has been the first large-scale conflict to demonstrate that commercial space systems can be directly integrated into military operations. In this context, the Starlink satellite internet system developed by SpaceX played a critical operational role in the initial stages of the war by ensuring the continuity of Ukraine's command and control (C2) infrastructure. Russia's targeting of Ukraine's terrestrial communications infrastructure in February 2022 exposed the fragility of traditional communications systems; the rapid deployment of Starlink terminals prevented the collapse of Ukraine's military and civilian communications networks (Boulder, 2025: 2-5).

Starlink's low-latency, high-bandwidth communication capabilities have become critical not only for high-level command elements but also for front-line troops and unmanned aerial vehicles. Thanks to this system, Ukrainian forces have been able to maintain real-time intelligence sharing, drone image transmission, and target coordination, thereby gaining operational agility (Watling & Reynolds, 2022, :18–22). Satellite-based communication has

significantly reduced the time required to adjust fire, particularly through the integration of artillery systems and drone reconnaissance. This situation demonstrated how space-enabled network-centric warfare works in practice.

However, the advantages of Starlink have also revealed significant operational vulnerability. Starlink is not a state-controlled military system, but a commercial infrastructure operated by a private company. This situation has shown that critical military communication infrastructure during war may depend on the technical, commercial, and political decisions of a non-state actor (Isaacson, 2023: 412). In particular, the limited coverage area of Starlink in certain regions or the restriction of some operational uses has revealed that the continuity of military operations may depend not only on technical factors but also on political and institutional decisions.

The fundamental operational lesson from this situation is that modern military forces must effectively use commercial space systems, but that complete dependence on them creates operational risks. According to NATO doctrine, command and control continuity is a fundamental element of operational success, and redundancy in communication systems is critical (NATO, 2019: 1-6). The example of Ukraine has shown that commercial space systems can play a key role in ensuring this continuity, but also that the lack of state control over these systems can create operational uncertainty.

Commercial Satellite ISR Capability: Maxar, Planet Labs, and ICEYE

The Ukraine–Russia war has demonstrated that commercial satellite intelligence (Intelligence, Surveillance, Reconnaissance – ISR) has become an integral part of modern warfare. High-resolution optical and synthetic aperture radar (SAR) imagery from Maxar Technologies, Planet Labs, and ICEYE has increased Ukraine’s operational situational awareness and accelerated decision-making on the battlefield. The buildup of Russian troops on the Ukrainian border in the early days of the war and the movement of forces along the front line in later stages were largely tracked by both Ukraine and the international community thanks to commercial satellite imagery (Boulton, 2022: 5–9; Defense Intelligence of Ukraine, 2023).

Maxar’s high-resolution optical imagery has enabled the analysis of force deployments at the strategic and operational levels (Maxar Technologies, 2022). Planet Labs’ satellite constellation, with its wide coverage area and high revisit frequency, has contributed to operational tempo analysis (Hamm, 2025). ICEYE’s SAR satellites have provided imagery independent

of weather conditions and nighttime conditions, producing critical intelligence, particularly during periods of intense conflict (ICEYE, 2022). This diversity has enabled Ukraine to establish a multi-layered ISR architecture and reduce its dependence on a single system.

From an operational perspective, commercial ISR systems have provided three key advantages. The first is accelerated decision-making. Near real-time image streaming has facilitated target identification and fire support planning (Watling & Reynolds, 2022: 24–30). The second is transparency and information superiority. Publicly sharing commercial imagery has limited Russia's information operations and facilitated the mobilization of international support. The third is cost-effectiveness. High-cost military reconnaissance systems that states do not possess have been procured from the commercial market.

However, commercial ISR systems also have some vulnerabilities. Since these systems are not under state control, continuity of access depends on political and economic factors. In addition, the resolution and revisit time of commercial imagery may be limited compared to military-grade systems. Electronic warfare and cyber threats can also target commercial space infrastructure (Secure World Foundation, 2024 :33). Therefore, commercial ISR capabilities cannot completely replace the national military space architecture; however, they serve as a force multiplier that complements it and provides operational flexibility.

The example of Ukraine demonstrates that space power no longer relies solely on state-owned military systems, but that the commercial space ecosystem can take on a direct operational role on the battlefield. This situation reveals that, within the concept of multi-domain operations, the space domain is being restructured through public-private sector integration.

Strategic Lessons

In the literature on warfare, “strategic lessons” are the outcomes drawn from the effects of a conflict on national objectives, the use of force, the architecture of deterrence, and long-term security policies, which shape states’ future strategic planning. According to Clausewitz, “*war is the continuation of politics by other means*,” and the strategic level refers to the direction of military means to serve political ends (Clausewitz, 1976: 87, 605). In this context, the strategic lesson is to understand the impact of war on the state’s grand strategy rather than its tactical successes or failures. Liddell Hart de-

defines strategy as “*the art of directing military means to achieve political ends*” (Hart, 1967: 335). Therefore, strategic lessons analyze how war transforms the state’s long-term security architecture, alliance relationships, and balance of power. Colin Gray emphasizes that strategy is a process requiring continuity and adaptation; strategic lessons form the basis for future deterrence and defense planning (Gray, 1999: 17).

The Russia–Ukraine war can be categorized in the literature as a “*regional limited war*” due to its structure, which involves two actors possessing nuclear weapons not engaging in direct conflict, remaining geographically confined to a specific region, and not escalating into a global war. The concept of limited war refers to conflicts in which the objectives, means, and geography are deliberately restricted (Clausewitz, 1976 :87; Freedman, 2017: 54). In such wars, the aim is not to destroy the opposing side, but to achieve specific political objectives using coercive force. Although Russia’s intervention in Ukraine initially aimed at regime change and geopolitical realignment, its avoidance of direct NATO intervention and the conflict remaining confined to Ukrainian territory prevented the war from escalating into a systemic global conflict. This situation can also be evaluated in the context of Schelling’s theories of compellence and deterrence; both Russia and NATO took care not to cross the strategic threshold, thereby exercising control over escalation (Schelling, 1966: 71).

In this context, space capabilities introduce a new dimension that contributes to maintaining strategic stability in a limited-warfare environment. During the Cold War, early warning systems and reconnaissance satellites were seen as key elements of nuclear deterrence (Gray, 1999: 152). Literature on the strategic use of space power has been influenced by the evolution of air power theory. Air power theorists such as Giulio Douhet and Billy Mitchell argued that a new domain (airspace) could change the strategic character of warfare (Douhet, 1983, p. 9, and Mitchell, 1942). Similarly, since the 1990s, space has increasingly been recognized as an independent strategic domain. One of the most influential figures in space power theory is Everett Dolman. According to Dolman, space, like sea lanes, is a “*high ground*” that can provide global dominance, and the state that gains superiority in space gains a decisive advantage in world politics (Dolman, 2002: 4). This approach can be seen as an adaptation of classical geopolitical theories to space. Colin Gray, on the other hand, evaluates space power in a strategic context, arguing that space systems change not nature but the character of

warfare (Gray, 1996: 311). According to Gray, space systems accelerate decision-making by providing information superiority, thereby strengthening strategic deterrence. The US Space Force doctrine also defines space as an independent warfare domain and views space superiority as a strategic objective (US Space Force, 2020: 5). Russia–Ukraine war provides practical validation of space power theory. Space-based communications, ISR, and navigation systems have produced not only tactical advantages but also strategic resilience. Space-enabled command and control continuity was decisive in preventing Ukraine from experiencing state collapse in the initial stages of the war. This demonstrates the capacity of space power to produce “*state resilience*” at the strategic level.

According to Mearsheimer’s realist balance-of-power approach, states seek to maximize their relative power to survive (Mearsheimer, 2001: 29). In the case of Ukraine, despite the direct military balance of power favoring Russia, space-enabled information superiority and allied integration partially offset this asymmetric balance. This situation demonstrates that the modern concept of power is shaped not only by conventional military capabilities but also by network-centric and space-enabled systems. Furthermore, space systems have enabled NATO’s indirect participation, strengthened alliance solidarity, and reshaped the deterrence architecture. When evaluated in the context of Schelling’s deterrence theory, space systems have provided transparency and early warning mechanisms that support escalation control (Schelling, 1966:71). In the Ukraine war, space systems have become not only a strategic early warning tool but also an element that enhances the operational effectiveness of conventional warfare. This situation validates the multi-domain warfare concept at the strategic level. (Burbach, 2022)

In summary, strategic lessons concern how space power transforms national and alliance-level strategies as well as the broader conduct of modern warfare.

-*Space as a strategic enabler*: Space-based intelligence (e.g., satellite imagery, communications, navigation) has proven vital to Ukraine’s national survival and Western support.

-*Resilience and deterrence*: The need for a resilient space infrastructure capable of withstanding cyberattacks, jamming, or anti-satellite operations.

-*Allied and commercial integration*: The strategic impact of private space companies (such as SpaceX’s Starlink) and cooperation between governments and the private sector.

-Information superiority: How space-based data shapes the strategic narrative — exposing Russian troop movements, countering disinformation, and mobilizing international support.

-Redefining deterrence and sovereignty: Space capabilities now play a role in national defense strategies, redefining how countries think about deterrence and escalation.

Based on this table, the strategic lessons can be summarized as follows:

Table -2 Strategic Lessons from the Russia–Ukraine War on the Role of Space Power (2022–Present)

Prior-ity	Strategic Lesson	Description	Evidence from Ukraine War	Strategic Im-plication
1	Space power enhances strategic resilience	Space-based com-munications and ISR enable states to maintain national and military func-tionality during large-scale attacks.	Ukraine maintained government, military, and national commu-nications via Starlink after Russian strikes on infrastructure (Erwin, 2022; Foust, 2022).	Space systems increase na-tional survival capacity and strategic endur-ance.
2	Commercial space actors have become strategic stake-holders	Private space com-panies now directly influence military operations and national security outcomes.	Starlink, Maxar, Planet Labs, and ICEYE provided essential wartime capabilities (Johnson, 2023; Sheetz, 2022, Albon,2022).	Strategic power is no longer monopolized by states; commercial integration is essential.
3	Space capabili-ties strengthen deterrence and international support	Space-based ISR enhances transpar-ency and facilitates international po-litical and military support.	Satellite imagery ex-posed Russian troop movements and supported NATO and allied response coordination (Ger-shenhorn & Sataria-no, 2022).	Space systems contribute to collective deterrence and alliance cohe-sion.
4	Access to space capabil-ity can com-pensate for conventional military asym-metry	States can offset conventional mil-itary inferiority through access to space-enabled capa-bilities.	Ukraine leveraged commercial and al-lied space systems to resist a convention-ally superior Russian military (Watling & Reynolds, 2022).	Space capabili-ty functions as a strategic force multiplier.

Prior- ity	Strategic Lesson	Description	Evidence from Ukraine War	Strategic Im- plication
5	Space has become a central domain in modern strategic competition	Control, access, and integration of space capabilities influence the overall strategic balance of power.	Russia targeted Ukrainian communications infrastructure and attempted to disrupt satellite networks (Secure World Foundation, 2023).	Space is now a decisive domain in modern warfare and national

So, strategic lessons are about how space power changes national and alliance-level strategy and the broader conduct of modern warfare.

Operational Lessons

In military warfare literature, “operational lessons” are conclusions drawn from experience gained by military forces during a conflict regarding operational planning, command and control, force coordination, and battlefield effectiveness, with the aim of enhancing the effectiveness of future operations. The operational level is a critical level that lies between strategic objectives and tactical applications and encompasses the planning, coordination, and execution of military operations (Joint Chiefs of Staff, 2017: I-5). In this context, operational lessons focus not only on the war’s outcome but also on how it was conducted and how military power was effectively used on the battlefield.

Soviet military thinkers developed the theoretical foundations of the operational level. In particular, Aleksandr Svechin argued that there was a separate operational level between strategy and tactics, and that the outcome of war was determined by planning and coordination at this level (Svechin, 1992: 68). According to Svechin, operational art is the process by which military forces are deployed and employed to achieve strategic objectives in a specific theater. Similarly, Soviet theorist Georgii Isserson emphasized that modern warfare requires continuous and deep operations and that superiority in information, movement, and coordination at the operational level is decisive (Isserson, /2013: 15). This theoretical approach formed the basis of modern multi-domain operations doctrine.

Western military theorists have also emphasized the importance of the

operational level. Martin van Creveld stated that command, communication, and information flow are the key determinants of operational success in modern warfare (Van Creveld, 1985: 269). According to him, the side that achieves information superiority on the battlefield gains an operational advantage thanks to its faster decision-making and implementation capacity. This approach later formed the theoretical basis for the concept of network-centric warfare. Arthur Cebrowski and John Garstka argued that information networks and sensor systems directly increase operational effectiveness and change not the nature of war, but the way war is conducted (Cebrowski & Garstka, 1998: 30).

The theoretical framework regarding the operational role of space capabilities has been developed primarily by Colin Gray and Everett Dolman. According to Gray, space systems play a significant role in achieving information superiority in modern warfare and in enhancing command-and-control, reconnaissance, and coordination capabilities at the operational level (Gray, 1996: 307). Gray states that space systems change the character of warfare by providing commanders with simultaneous situational awareness. Everett Dolman argues that space systems significantly increase the effectiveness of conventional forces by improving coordination and operational reach on the battlefield (Dolman, 2002: 87).

The US military doctrine also defines space systems as a critical force multiplier at the operational level. According to the US Space Force doctrine, space systems increase operational effectiveness and enable forces to move in a synchronized manner thanks to the support they provide in the areas of command and control, intelligence, navigation, and communications (US Space Force, 2020:10). These systems enable the integration of land, air, sea, cyber, and space elements, particularly within the concept of multi-domain operations.

Based on concrete cases from the 2022 Ukraine war, operational lessons regarding space capabilities can be summarized in order of importance (starting with the most critical) as follows:

1. Command and Control Continuity is a Prerequisite for Operational Survival (Starlink Case – February 2022)

In February 2022, Russia conducted a cyberattack on Viasat in the first hours of the invasion, affecting the communications infrastructure. Targeting Ukraine's terrestrial communications infrastructure and the Viasat KA-

SAT system demonstrated that Command, Control, and C2 infrastructure is a priority target in modern warfare. Elon Musk's Starlink terminals were sent to Ukraine. Frontline troops, drone operators, and artillery units coordinated via satellite internet. Ukraine's rapid deployment of Starlink terminals ensured continuity of communication for the state and the military, preventing operational collapse (Rodin, Holynska, Tretter, Van Bibber; 2025:8-14).

Operational Lesson:

Distributed, low-orbit satellite communication systems (LEO) are more resilient than traditional communication infrastructure in high-intensity combat environments. If C2 continuity cannot be maintained, operational integrity collapses.

2. Space-Based ISR Determines Operational Tempo and Targeting Time (Maxar – Planet Labs – ICEYE Cases)

Commercial satellite companies enabled Ukraine to monitor Russian troop movements by providing high-resolution optical and SAR imagery. In February 2022, Maxar imagery revealed Russian troop buildups around Kyiv. Planet Labs monitored changes along the front lines with their daily revisit capability. ICEYE SAR satellites provided imagery even in cloudy weather conditions.

Operational Lesson:

Near real-time commercial ISR data accelerates the decision-making cycle (OODA loop) and shortens targeting processes. Operational agility depends on information superiority.

3. Space-UAV-Artillery Integration Creates a Force Multiplier Effect (HIMARS – Drone Coordination, Summer 2022)

Satellite communication and GPS-aided targeting enabled Ukraine to use its HIMARS systems with high precision. In Summer 2022, HIMARS systems struck Russian ammunition depots. Drone reconnaissance data was transmitted via satellite communications. GPS-based coordinates enabled precision strikes (Kozatskyi, 2026).

Operational Lesson:

Space systems alone are not decisive; however, when integrated with tactical systems, they can shift the operational balance.

4. Dependence on Commercial Space Actors Creates Strategic Risk (Starlink Crimea Crisis – End of 2022)

Starlink’s service restrictions in certain regions have shown that military operations may depend on the political decisions of a private company. In late 2022, allegations emerged that SpaceX had restricted access around Crimea. This situation revealed that private company control could create operational risk (BBC News Russian, 2026).

Operational Lesson:

Commercial systems offer flexibility; however, operational reliance is risky without government oversight and alternative systems.

5. GPS and PNT Systems Are Vulnerable to Electronic Warfare (Russian GPS Jamming Activities)

Russia has conducted GPS jamming activities, particularly along the front lines. Intensive electronic warfare applications in the Donbas region. Navigation disruptions in drone and munitions systems (Business Insider, 2023).

Operational Lesson:

Space-based PNT systems are vulnerable without anti-jamming capabilities and alternative navigation systems.

6. Transparency and Information Superiority Produce Strategic Impact (Release of Maxar Images to the Public)

The public release of commercial satellite images has limited Russia’s information operations. Kyiv convoy images. These images are related to the Bucha region (BBC, 2022).

Operational Lesson:

Space systems produce operational and strategic impact not only in a military context but also in information warfare.

7. Distributed LEO Architecture Increases Resilience

Starlink’s structure, based on thousands of small satellites, has made it difficult to completely disable the system.

Operational Lesson:

A distributed satellite architecture is more resilient than a single, large platform.

Table 3 Operational Lessons from the Russia–Ukraine War on the Military Use of Space (2022–Present)

1	Space-based communications ensure C2 continuity	Satellite communications maintain command and control when terrestrial infrastructure is disrupted.	Starlink enabled Ukrainian military and government communications after Russian strikes on infrastructure; (-Foust, 2022).	Prevented operational paralysis; sustained command continuity.
2	Commercial satellite ISR enhances situational awareness	Commercial imagery provides near-real-time intelligence on adversary movements.	Maxar and Planet Labs imagery revealed Russian troop buildup prior to invasion (Sheetz, 2022).	Improved operational planning and targeting accuracy.
3	Space-enabled precision targeting integration	Satellite ISR and GNSS data integrate with artillery and drone systems.	Satellite data supported HIMARS and drone-based targeting operations (Watling & Reynolds, 2022).	Increased strike precision and reduced ammunition expenditure.
4	Distributed LEO constellations increase resilience	Large commercial constellations are harder to disrupt than single military satellites.	Russian jamming attempts failed to significantly disrupt Starlink operations	Ensured redundancy and operational continuity.
5	Rapid commercial access accelerates adaptation	Commercial providers deliver capability faster than traditional procurement systems.	Ukraine received Starlink terminals within days of request (Foust, 2022).	Enabled rapid operational recovery during crisis.
6	Space supports decentralized warfare	Satellite communications enable distributed small-unit operations.	Ukrainian drone units transmitted targeting data via satellite links (Watling & Reynolds, 2022).	Increased operational flexibility and tempo.
7	Commercial space compensates for limited sovereign capability	States without strong national space systems can leverage commercial access.	Ukraine relied on Starlink and commercial ISR instead of national military satellites (Johnson, 2023).	Functioned as a force multiplier.
8	Dependence on commercial providers creates vulnerability	Corporate decision-making can affect military operations.	SpaceX restricted Starlink use for certain offensive operations	Revealed operational risk of private-sector dependence.
9	Space-based ISR enhances information warfare	Satellite imagery counters disinformation and supports strategic communication.	Commercial imagery publicly documented battlefield developments	Strengthened international legitimacy and support.
10	Space integration accelerates decision cycles	Real-time data shortens observe–orient–decide–act (OODA) loops.	Rapid targeting updates supported adaptive Ukrainian responses.	Improved opera

As shown in the table above, the operational lessons from the Russia–Ukraine War (2022–present) demonstrate that space capabilities have become integral to modern military effectiveness. The conflict illustrates that space-based communications, particularly through systems such as Starlink, ensured continuity of command and control (C2) despite sustained attacks on terrestrial infrastructure. At the same time, commercial satellite imagery providers—including Maxar, Planet Labs, and ICEYE—significantly enhanced situational awareness, targeting accuracy, and operational tempo. The integration of satellite-enabled ISR with precision fires, drones, and de-

centralized units accelerated decision cycles and increased battlefield adaptability. However, the war also revealed operational vulnerabilities associated with dependence on commercial providers, highlighting risks related to access control and corporate decision-making. Overall, the evidence suggests that distributed commercial space architectures functioned as force multipliers for Ukraine, enabling resilience, agility, and sustained operational performance in a contested environment.

Conclusion

The Russia–Ukraine war has clearly demonstrated that space power is a decisive factor at both the operational and strategic levels of modern warfare. The primary research question addressed in this study concerns how the Russia–Ukraine war reveals the role of space capabilities in operational effectiveness and strategic resilience, and how this influences modern warfare doctrines. The findings indicate that space systems significantly enhanced Ukraine’s warfighting capabilities, playing a critical role in maintaining command-and-control continuity, enhancing situational awareness, and enabling precision targeting. Despite Russia targeting Ukraine’s communications infrastructure in the initial stages of the war, the deployment of the Starlink satellite communications system ensured the continuity of Ukraine’s military and political decision-making processes and prevented operational collapse. Similarly, commercial satellite intelligence from Maxar Technologies, Planet Labs, and ICEYE enabled Ukraine to track Russian troop movements, accelerate targeting processes, and control operational tempo. While space systems did not grant Ukraine absolute military superiority, they directly influenced the course of the war by enhancing operational resilience and sustainability.

The most important operational lesson is that space-based communication and intelligence systems underpin command-and-control continuity in modern warfare. This demonstrates that information, coordination, and operational synchronization, as emphasized in Svechin and Isserson’s theory of operational art, can be achieved through space systems. The case of Ukraine illustrates that space systems serve as significant force multipliers, particularly when integrated with drones, artillery, and ground forces. However, the war has also revealed that dependence on space systems carries inherent risks. Specifically, the control of commercial space systems by private companies has emerged as a factor that could create strategic and operational fragility

regarding the continuity of military operations. This underscores that space power in modern warfare is not only a technical capability but also part of a broader political, economic, and institutional ecosystem.

At the strategic level, the war in Ukraine has demonstrated that space power directly affects states' strategic resilience and deterrence capabilities. Consistent with Clausewitz's view that war is a means to achieve political ends, space systems have prevented strategic collapse by ensuring the sustainability of Ukraine's warfighting capabilities. Additionally, space systems have enabled NATO to support Ukraine without direct engagement, highlighting space as a critical component of modern deterrence architecture. This outcome supports Gray and Dolman's theoretical perspective that space power is a domain conferring strategic superiority. The war in Ukraine has shown that, while space systems have not fundamentally changed the nature of war, they have significantly transformed its character, speed, and mode of execution.

Another research question addressed in this study concerns how the role of commercial space actors on the battlefield affects the use of military force. The case of Ukraine demonstrates that commercial space companies have become direct operational actors in modern warfare. Starlink's provision of communications infrastructure and the imaging support from commercial ISR companies reveal that states can now build operational space power not only through national capabilities but also by integrating into the commercial space ecosystem. This development has led to a redefinition of space power and indicates that public-private sector integration in the space domain is now a fundamental element of modern warfare.

Considering these findings, the central thesis of this study—that space power has become a decisive force multiplier in modern warfare—has been confirmed. The war in Ukraine demonstrates that space systems increase operational effectiveness, strengthen strategic resilience, and can influence the duration and outcome of conflict. (How the war in Ukraine is affecting space activities, n.d.) However, the conflict also shows that space power alone does not guarantee victory; it is most effective when combined with conventional military power, economic capacity, and political support.

This war offers important insights for NATO member states and non-NATO countries that perceive Russia as a potential threat. First, these countries should develop space-based communication and intelligence capabilities

and integrate these systems with their military forces. Distributed satellite architectures, anti-jamming technologies, and redundant communication systems are critical for ensuring operational continuity in modern warfare. Second, states should establish institutional mechanisms for cooperation with commercial space companies to ensure continuous access during crises. Third, protecting space systems against electronic warfare and cyber-attacks must become a fundamental element of military planning. Finally, integrating space systems with land, air, sea, and cyber forces is essential for the effective implementation of the modern multi-domain operations concept. Space systems have become an integral component of modern warfare, and the capacity to access, effectively utilize, and protect space systems is a fundamental determinant of modern military power. For states seeking to deter actors with high conventional military capabilities, such as Russia, developing space power is no longer an option but a strategic necessity. This war has clearly demonstrated that space is one of the fundamental domains shaping the operational effectiveness and strategic outcome of modern warfare.

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